

Why Few and Why Here? Specialist Supply and Location Choice in Colombia

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“During the process, participants experienced a range of negative emotions, most often frustration, anger, anxiety, uncertainty, sadness and fear. These experiences led them down two different paths. On one side, 13 participants (59 %) hold on to their decision to specialize, although they have begun to consider alternatives such as migration. In fact, 63.6 % have considered specializing abroad, mainly in Spain, Argentina, the United States, Mexico and Brazil, motivated by the perception that selection processes there are fairer, slots are more available and quality of life is better. This matters all the more given that for 72.7 % of participants medical residency remains a priority.”

Martínez-Forero et al. (2025), my translation

Semana

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OPINIÓN

¡Ojo! Los médicos colombianos se están yendo del país

Conseguir recursos requiere voluntad y decisiones; formar un especialista toma años.

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Cristina Plazas Michelsen

27 de septiembre de 2026 a las 5:00 a. m.



EL ESPECTADOR



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Necesitamos más médicos especialistas, pero escasean los cupos en los hospitales

Algunas universidades señalan que este semestre académico han tenido dificultades para garantizar un cupo de práctica médica en hospitales a sus estudiantes de pregrado de medicina y de residencia. Los alumnos señalan que eso está perjudicando su formación. ¿Qué está pasando con la formación de los futuros doctores?



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Paula Casas Mogollón

04 de noviembre de 2025 - 08:00 a. m.

[more press](#) →

Colombia has few specialists relative to the OECD

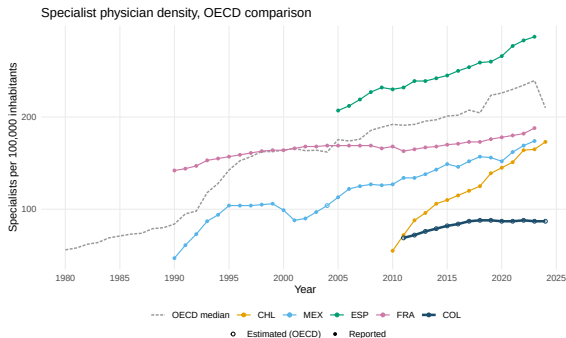


Figure: Specialists (ISCO-08 group 221) per 100 000 inhabitants. OECD median over countries with data in each year. Source: OECD Health Statistics (December 2025).

Where specialists are fewer, treatable deaths are higher

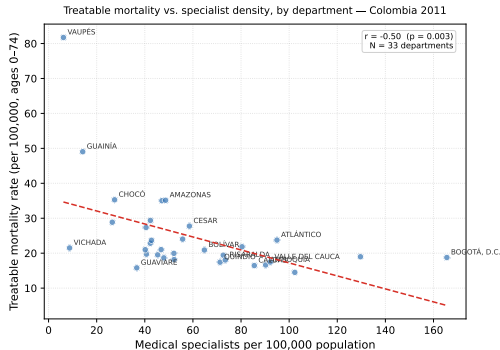


Figure: Departments, 2011. Dashed line: OLS fit. Source: DANE EEVV, ReTHUS/PILA.

[other years →](#)

[time series →](#)

[what is treatable? →](#)

Specialists earn a large premium

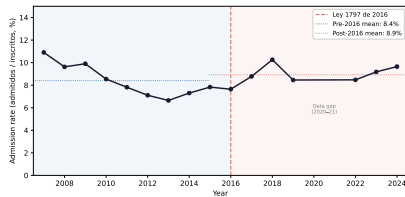
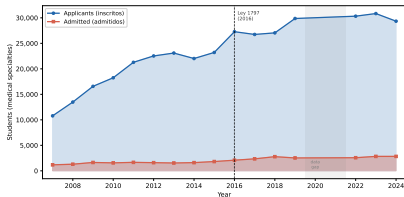
	log(real hourly earnings)	
	(1)	(2)
Specialist	0.597*** (0.017)	0.501*** (0.017)
Age		0.061*** (0.005)
Age ²		-0.0006*** (0.0001)
Year fixed effects	Yes	Yes
Observations	5 772	5 772
R ²	0.177	0.221

General practitioners and specialists, GEIH 2022–2025 (DANE).
 OLS weighted with expansion factors. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

At the same age, a specialist earns $\approx 65\%$ more per hour than a general practitioner ($e^{0.501} = 1.65$).

dispersion →

But supply does not respond: admission is rationed



Admitted / applicants, 2007–2024 (excl. 2020–21).

Source: SNIES.

- Earnings rise; graduates grow slowly.
- Only 1 in 11 applicants is admitted, unchanged since 2007.

numbers →

The press suspects, but there is no verdict

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- **Suspicion:** a “**specialists’ cartel**”; 32 % of hospital managers say specialists control the number of graduates (“Déficit de especialistas, ¿quién tiene la culpa?”, 2013); in applicants’ accounts, “**most slots are handed out by connections**” (*a dedo*) (Correa, 2013).

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- **Dispute:** Ascofame speaks of “isolated cases” (Moreno Agudelo, 2013).
- **Since 2025:** training sites are scarce (Casas Mogollón, 2025).

Quotes translated from Spanish.

coding of 47 articles →

Why does specialist supply not respond?

H1: entry restriction

Incumbent specialists limit slots to protect their rents.

H2: capacity

Training sites, faculty and hospitals are lacking.

H2a: slots are unprofitable to expand under tuition caps and thin per-trainee financing. [predictions →](#)

Both predict few graduates and high earnings, but call for **opposite policies**.

This paper:

- 1 Links ReTHUS, PILA and SNIES to follow the chain slot → graduate → workplace.
- 2 Uses Ley 1797 of 2016 as a change in slot regulation.
- 3 Estimates where new specialists choose to work.

Contribution

- 1 Health economics of physician supply.** Micro evidence on what holds back specialist training in a middle-income country (Bardey, 2014; Kessel, 1958; Natalia, 2014). New linkage ReTHUS–PILA–SNIES.
- 2 Entry regulation in professions.** Confronts entry restriction with capacity constraints where barriers are informal as well as formal (Schaumans & Verboven, 2008; Shaked & Sutton, 1981); documents a reform that was never implemented and what its null can and cannot tell us.
- 3 Market structure and health outcomes.** Relates specialist maldistribution to treatable mortality across departments, where high-income evidence is mixed (Baicker & Chandra, 2004; Starfield et al., 2005) and Latin American evidence is scarce (Mullachery et al., 2022).

Who decides how many slots there are?

- 3–4 year residency in a **university program**, not a hospital.
- **MEN** grants the *registro calificado* and sets maximum capacity.
- **MinSalud** regulates training sites (*escenarios de práctica*).
- **Universities** decide how many to admit each year.

There is no national inventory of slots: the decision is **decentralized and opaque**.

[pipeline detail →](#)[university hospitals →](#)[programs by department →](#)

Ley 1797 of 2016, Article 23

- Institutions with high-quality accreditation (CNA) *may* (*podrán*) expand slots, with approval from MinSalud and MEN.
- The article was **never regulated**: the pathway exists in the law but has no procedure.

What we estimate

The effect of **being eligible** (accredited in 2016) on new entrants: an intention-to-treat effect.

Data

- **ReTHUS:** health workforce registry; specialty, institution and year of title. Measures the stock of specialists. [stock →](#)
- **SNIES:** applicants, admitted and graduates by program, 2007–2024.
- **PILA:** municipality where each specialist works (formal employment), 2011–2019.
- **GEIH:** physician earnings, 2008–2025.

Design: one eligible cohort, not-yet-eligible controls

- **Eligible:** the 5 institutions already accredited when the law passed (2016 cohort).
- **Controls:** 20 institutions accredited between 2017 and 2025, used only while not yet eligible: the pool shrinks from 20 (2017) to 12 (2022).
- **Outcome:** new entrants per institution and year, 2017–2022.
- **Estimator:** Callaway and Sant’Anna (2021), which handles the shrinking control pool; Borusyak et al. (2024) imputation as robustness.

$$ATT_{2016} = \mathbb{E} [Y_t(2016) - Y_t(\infty) \mid G = 2016]$$

ATT_{2016} : effect of **being eligible** (ITT), not of using the pathway.

If the law had relaxed a binding constraint, we would expect

$$ATT_{2016} > 0.$$

Eligibility did not increase new entrants

Sample	(1) CS-DiD All	(2) PPML All	(3) PPML High rejection	(4) PPML Low rejection
Eligible $\times$ post	-14.46 (11.22)	-0.183 (0.145)	-0.241* (0.128)	0.053 (0.282)
95 % CI	[-32.7, 3.7]			
Counterfactual mean	117.9			
Institution $\times$ specialty FE		Yes	Yes	Yes
Year FE		Yes	Yes	Yes
Observations	400	6 219	4 294	1 821
Institutions	25	25	24	17
of which eligible	5	5	5	5

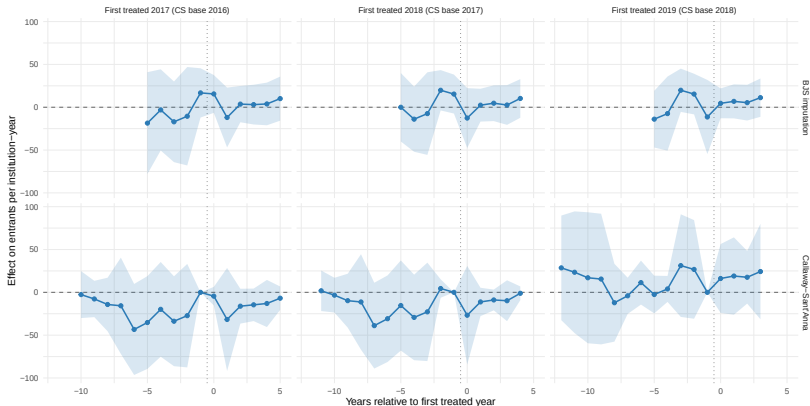
Outcome: new entrants (SNIES). (1) levels by institution-year; SE and CI by multiplier bootstrap (9 999 replications). (2)–(4) Poisson by institution-specialty-year; SE clustered by institution. High/low rejection: specialty's pre-2016 rejection rate above/below the median. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

No detectable effect, but with **only 5 eligible institutions** the 80 % MDE is $\approx 23\%$ of the counterfactual, and (1) leans on 2016 as its base year.

base year →

The sign of the estimate depends on the base year

Ley 1797, 2016 cohort: event studies by treatment timing and estimator



CS differences every post year against one base year; 2016 was high for the eligible 5, so the CS estimate moves from -14.5 to $+19.3$ as the base shifts. BJS uses all untreated years and stays near zero: $+1.5$ to $+7.0$, $SE \approx 10$.

[table →](#)

Where do new specialists choose to work?

	Conditional logit
log(municipal wage)	3.528*** (0.128)
Municipal HHI	-4.643*** (0.080)
log(n. specialists)	0.661*** (0.004)
Trained here	-0.814*** (0.047)
log(1 + distance)	-0.624*** (0.008)
Specialists	45 590
Municipalities in choice set	100
Log-likelihood	-101 321

PILA 2011–2019, first observed municipality of work. BHHH
SE. *** $p < 0.01$.

- They go to municipalities with **high wages** and **many colleagues**.
- They avoid **concentrated** markets.
- Distance matters.

Conditional on size and distance, staying where they trained is **less likely**: consistent with entry frictions, but also with saturation.

sample →

Conclusions and next steps

What we know

- There are rents: $\approx 65\%$ hourly premium at equal age.
- There is rationing: 1 in 11 applicants gets in.
- Eligibility under Ley 1797 did not produce more entrants; since the article was never regulated, this null alone does not separate H1 from H2.

What comes next

- Heterogeneity by incumbent density in each specialty.
- Logit: use the undergraduate municipality as origin to separate entry frictions from returning home.
- Counterfactuals: more slots and their effect on treatable mortality.

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Thank you!



Working paper (SSRN)



Personal website



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Víctimas de las roscas

Cada médico que ha intentado especializarse tiene una historia sobre lo mal que funcionan las cosas. Todos los relatos coinciden: la mayoría de los cupos se asignan a dedo.

Pablo Correa

12 de octubre de 2013 - 04:00 a. m.

El Espectador, Correa (2013)



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Universidad Colombiana
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Colombia: Con pocos especialistas en salud, y es un drama poder estudiar una especialización

Sept 5/21 Son escasos los especialistas, y ganan muy bien, mientras que miles de profesionales, que ganan mal y están dispuestos a pagar por formarse, ruegan por lograr uno de los muy pocos cupos. ¿Por qué?

El Observatorio de la Universidad Colombiana (2021)

ediciónmédica

1508

INICIO

SALUD PÚBLICA

PROFESSIONALS

GESTIÓN

FMPRES

PROFESIONALES

LLAMADO A MINSALUD

ANIR: en Colombia no es fácil acceder a un cupo para adelantar una especialización en Medicina por falta de cupos

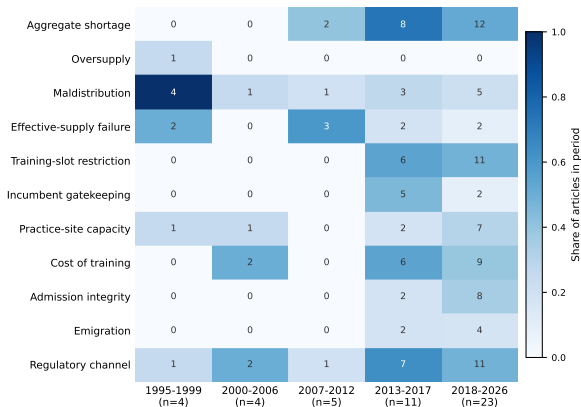
En el país hay por lo menos 400 especialidades médicas ofrecidas por las facultades de medicina en todo el territorio

Martes, 13 de septiembre de 2022, a las 16:06

Edición Médica, Murillo (2022)

← back

Diagnostic frames in the press, by period



Number of articles by frame and period; purposive sample of 47 press articles and one academic article; LLM coding (pooled $\kappa = 0.72$). See thesis appendix.

← back

Treatable mortality falls, but the gap persists

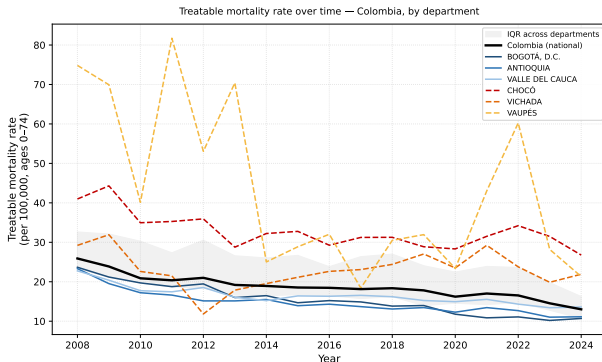


Figure: Treatable mortality (per 100 000, ages 0–74) by department, 2008–2024. Band: interquartile range. Blue: high-specialist departments; red/orange: low-specialist departments. Source: DANE EEVV.

Treatable vs. preventable mortality

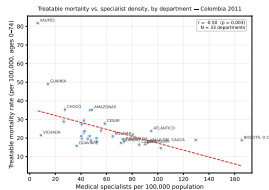
- **Preventable:** avoidable through public health (vaccines, tobacco, road safety).
- **Treatable:** avoidable through timely and effective medical care.
- Only deaths before age 75 (Nolte & McKee, 2004).

We use the OECD/Eurostat list of treatable ICD-10 codes (OECD & Eurostat, 2022): ischemic heart disease, stroke, appendicitis, breast and cervical cancer, among others.

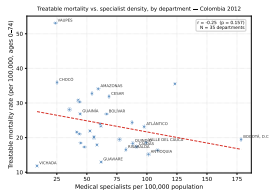
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Treatable mortality vs. specialist density, by year

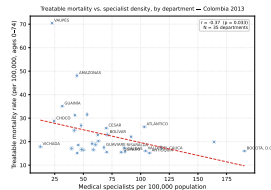
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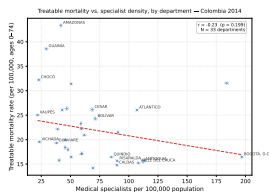
2011



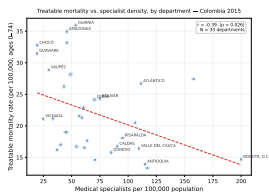
2012



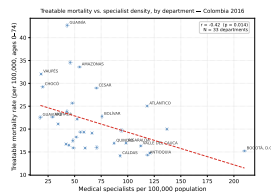
2013



2014



2015



2016



Health is an internally polarized sector

Earnings dispersion (GEIH 2024)

- Coefficient of variation of earnings in health: **1.335**, among the highest across sectors.
- 8.4 % of formal health employment earns more than twice the sectoral mean.

Consistent with Fuchs (1972) (U.S., 1959), though health does not rank first here as it does there.

Physician earnings, 2022–2025

Group	Mean monthly earnings (COP million)
Specialists	10.63
General practitioners	5.01
Other health workers	1.38

Raw monthly ratio, specialists / GPs: **2.12**.

Source: GEIH 2022–2025, DANE.

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Admission funnel: numbers

Table: Enrollment funnel for medical-surgical specialization programs, Colombia 2016–2024.

Year	Inscritos	Admitidos	Primer curso	Acceptance	Yield
2016	27,280	2,144	1,914	7.9%	89.3%
2017	26,726	2,379	1,918	8.9%	80.6%
2018	26,795	2,760	1,912	10.3%	69.3%
2019	29,591	2,523	2,038	8.5%	80.8%
2020	32,931	2,255	1,912	6.8%	84.8%
2021	33,130	3,008	2,119	9.1%	70.4%
2022	30,663	2,607	2,424	8.5%	93.0%
2023	31,075	2,873	2,343	9.2%	81.6%
2024	29,644	2,839	2,662	9.6%	93.8%

Notes: Acceptance = Admitidos / Inscritos.

Yield = Primer curso / Admitidos. Source:

SNIES, Ministerio de Educación Nacional.

H2a: financial constraints on training programs

- The ceiling sits in **program economics**, not physical inputs: tuition caps, thin per-trainee subsidies, capital-market frictions.
- **Predicts:** faster expansion in private programs; slot growth correlated with subsidy generosity across programs.
- **Weakened if:** public and private programs expand at similar rates, or slots are uncorrelated with per-trainee financing.

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Governance of residency slots: detail

The training pipeline

- 1 **6-year MD** (undergraduate)
- 2 **3–4 year residency** in a university program
- 3 Specialist title: practice begins

Before 2018 residents paid tuition and worked unpaid. **Ley 1917 of 2018** introduced a stipend, but does *not* affect slot numbers.

Who controls what

- **MEN:** *registro calificado* (mandatory license); maximum program capacity, not annual admissions.
- **CNA:** voluntary high-quality accreditation; no slot-setting role.
- **MinSalud / CITHS:** training-site capacity; no central planning mechanism.
- **Universities** ($\approx 65\%$ private): annual admissions within MEN-approved capacity.

← back

University hospital vs. training site

- **Hospital universitario** is a demanding legal category (accreditation, research, teaching). From 2027 only compliant institutions may use the name (Ley 2333 of 2023).
- But **any IPS** with a teaching-service agreement (*convenio docencia-servicio*) can host residents (Decreto 2376 of 2010; Ley 1872 of 2016).

For H2, the relevant constraint is how many IPS have agreements, services and faculty; MinSalud's list of IPS approved as training sites (September 2025) is its best proxy.

← back

Training is concentrated where specialists already are

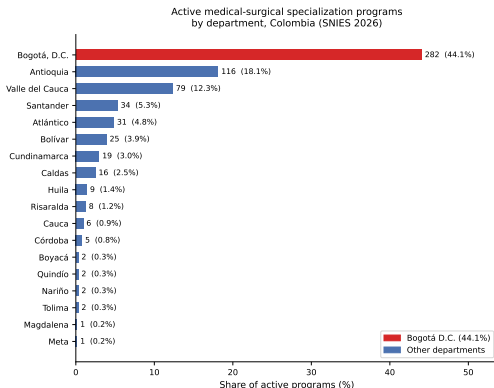
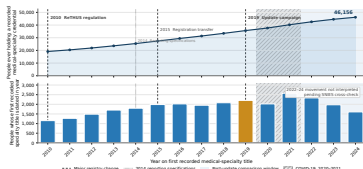


Figure: Active medical-surgical specialty programs by department: 640 programs in 18 departments and the Distrito Capital. Source: SNIES, MEN (March 2026).

Stock of specialists in ReTHUS

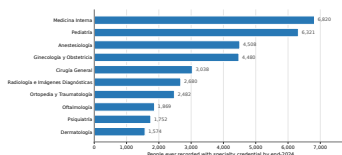
ReTHUS registry stock and first specialty-title years, 2010-2024
A cumulative credential registry reconstructed from the 2026 snapshot – not active specialist supply



Source: Ministerio de Salud y Protección Social, ReTHUS snapshot dated 2026-10-01.

Notes: Stock includes people ever observed with a qualifying credential through any end-point; results are unrounded. ReTHUS is real registration year, so later facilities can populate earlier cohorts.

Largest recorded medical-specialty stocks in ReTHUS, 2024
Cumulative credentials, not currently active clinicians



Source: Ministerio de Salud y Protección Social, ReTHUS snapshot dated 2026-10-01.

Notes: Specialists are non-exclusive, so a person with multiple credentials can appear in multiple lists; unspecified 'other specialty' categories are omitted.

Left: stock and titles by year. Right: stock by specialty, 2024. Source: ReTHUS.

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Ley 1797: timing and estimator robustness

First treated year	2017	2018	2019
CS base year	2016	2017	2018
Post window	2017–2022	2018–2022	2019–2022
Callaway-Sant’Anna	-14.46 (11.22)	-11.56 (11.36)	19.26 (22.27)
% of counterfactual	-12.3	-10.2	22.2
mean pre-period ATT	-22.2	-15.5	12.6
BJS imputation	4.03 (10.53)	1.47 (10.02)	6.99 (9.50)
% of counterfactual	4.0	1.5	7.1
Min. not-yet-eligible controls	12	14	15

Outcome: new entrants per institution-year, 2016 cohort; later cohorts are controls only while not yet eligible. Each column shifts the first treated year for every cohort (one or two extra years of approval lag). CS: long difference from the base year, multiplier-bootstrap SE. BJS: institution and year effects fitted on all untreated institution-years; SE clustered by institution, approximate with 5 treated units. % of counterfactual: estimate over observed treated mean minus estimate.

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Conditional logit: sample and specification

- **PILA 2011–2019:** municipality of work of formally employed specialists.
- **ReTHUS:** municipality of the training institution.
- **DANE MGN 2018:** municipal centroids for distances.
- Choice set: 100 municipalities with ≥ 5 specialists.
- 45 590 specialists; 11 530 excluded (trained abroad or missing code).

$$V_{ij} = \alpha \log(w_j) + \beta \text{HHI}_j + \gamma \log(n_j^{\text{spec}}) + \delta 1[\text{trained here}]_{ij} + \varphi \log(1 + \text{dist}_{ij})$$

Since $\log(1 + 0) = 0$, staying is still the option with the highest net utility: against a municipality 100 km away, $-0.81 + 0.62 \cdot \ln 101 \approx +2.1$. $\delta < 0$ says they stay *less* than size and distance predict.

← back

Model, stage 1: slots and entry restriction

$$K_{srt} = \min \left\{ \underbrace{\bar{C}_{srt}}_{\text{physical capacity}}, \underbrace{\bar{C}_{srt} - \phi G_{srt}}_{\text{permitted capacity}} \right\}$$

- K_{srt} : slots in specialty s , institution r , year t .
- G_{srt} : incumbent influence over the program.
- $\phi \geq 0$: intensity of entry restriction.

H2 dominates if the first constraint binds; H1 if the second does.

Total entrants $M_{st} = \sum_r K_{srt}$ set the size of the pool in stage 2.

stage 2 →

Model, stage 2: location choice

Each of the M_{st} graduates chooses municipality j to maximize

$$U_{ij} = V_{ij} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \text{i.i.d. type I extreme value}$$

with V_{ij} as in the estimated conditional logit [spec →](#).

Joining the stages

Stage 1 says *how many* specialists are missing if $\phi > 0$. Stage 2 simulates *where* they would have gone, and with the mortality gradient, what that is worth.

[← stage 1](#)